

GDGOC MODULE 2: VEHICLE FUEL EFFICIENCY (MPG) ANALYSIS REPORT PROJECT OVERVIEW

Subject: Vehicle Fuel Efficiency (MPG) Analysis

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Date: July 17, 2026

Today i started working on module 2 which is about car engineering data. The goal is to analyze what makes a car fuel-efficient using the seaborn mpg dataset. I am using numpy for the heavy Math, like solving linear regression and pca from scratch, and seaborn for the visuals.

Stage 1: Inspection and Cleaning

I loaded the dataset and saw 398 rows and 9 columns. I checked for missing values and found 6 empty spots in the horsepower column. I decided to use median imputation to fill them. I chose the median because it's safer than the mean if there are any extreme car engines that could skew the average.

Then i built a summary table using only numpy. I had to calculate the mean, median, and iqr manually. I noticed that mpg has a mean of 23.51 but a median of 23.0, which suggests the data is a bit skewed.

Stage 2: Statistical Analysis

In this stage, i extraacted the numeric specs like horsepower and weight into a numpy array. I performed z-score standardization manually using broadcasting. It's cool how $(x - \text{mean}) / \text{std}$ works for the whole matrix at once in numpy. My correlation matrix showed that weight has a vry strong negative relationship with mpg (-0.83). I also used boolean masking and found that cars with above-average horsepower are about 845.07 lbs heavier than the dataset average. This confirms that bigger engines usually mean heavier cars.

Stage 3: Visualization Journey

I made four plots to see if the math matches the pictures.

- Mpg distribution: the histogram shows a right-skew, meaning most cars have low mpg but a few are super efficient.
- Origin comparison: japan builds the most efficient cars in this dataset, while the usa has the lowest median mpg.
- Weight vs mpg: the scatter plot with my manual trend line shows a clear downward slope.
- Heatmap: it's very obvious that weight, displacement, and horsepower are all highly correlated, which is called multicollinearity.

Stage 4: Contextual Interpretation

Based on everything, weight is the main predictor of fuel efficiency. Heavier cars need more energy to move, so their mpg drops. I also grouped the data by decade and saw a big jump in mpg from the 1970s to the 1980s. I learned that the oil crisis in the 70s forced engineers to build smaller, more aerodynamic cars.

Bonus 1: Normal Equation Regression

I implemented linear regression without using sklearn. I used the normal equation formula with matrix inversion. I had to add a "bias column" of 1s so the model could find the intercept. My manual results (slope: -0.0077) matched numpy's polyfit exactly, which proves my matrix math is correct. My model has an error (rmse) of about 4.33 mpg.

Bonus 2: PCA From Scratch

I did principal component analysis using eigenvalues and eigenvectors. I reduced 5 features down to just 2 dimensions. Pc1 alone explains 78.4% of the variance. In my final plot, you can clearly see cars from the usa, japan, and europe clustering in different spots, which shows they have different "design signatures".

Bonus 3: Reusable datasetprofiler class

I refactored my code into a python class called datasetprofile. It has methods for stats, correlation, and a 4-subplot dashboard. I used type hints to make it professional. I used an AI tool to generate the docstrings, which saved a lot of time and made the code easier to read for other engineers.

Final Reflections

This module taught me how to bridge the gap between abstract math and real engineering questions. Using AI as a partner helped me fix a few small logic bugs and write better documentation. I feel more confident building my own machine learning tools from scratch now (i think).